Lesson plan · for the teacher

Good data, fair AI

AI · Level 1 · Lesson 3

Time
40 min

Materials
Board and chalk (or paper on the wall); Scrap paper and pencils; Two or three local fruits, or drawings of them; Worksheet: Spot the data gap (one per pair, optional)

Learning goals

- Explain that a machine that learns from examples can make more mistakes when some examples are missing.
- Spot who or what is missing from a set of examples.
- Suggest a fair way to fill a data gap, such as adding examples from the people left out.

Activity: Whose examples?: Test it

1. Warm-up (5 min): Ask: has a phone or app ever got someone's name or words wrong? Take 3–4 stories. Write "Why?" on the board.
2. Draw examples (8 min): Each pupil draws one fruit on scrap paper and names it. Put all the drawings in the "Our examples" box. Count each kind together.
3. Test it (10 min): The class is the recogniser and may only answer with a fruit in the box. Show a common fruit, then a local fruit nobody drew. Pupils vote for the closest drawing.
4. Pair talk (10 min): Pairs discuss: what was missing, and how could we fix it? Then ask: if voice typing learned mostly from some accents, whose voices might it get wrong?
5. Wrap-up (7 min): Add drawings of the missing fruit and test again. Write on the board: "It's not just how many examples — it's whose." Then do the exit ticket.

Draw a big box on the board labelled "Our examples". Bring two or three local fruits pupils may not think to draw, such as guava, dates, prickly pear, jackfruit or a baobab pod.

Discussion

- Which drawing on the board is closest to the new fruit?
- Is that the right answer?
- So how could we make our examples fairer?

Exit ticket

- One thing missing from our examples, and one way to fix it.

Try it in class tomorrow
Play "Whose examples?": Pupils draw one fruit each and stick the drawings on the board — these are the examples. The class is a "fruit recogniser" that can only answer with fruit on the board. Hold up a local fruit nobody drew. Ask: what went wrong, what was missing, and how is this like voice typing and accents?

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Voice typing makes more mistakes for Salma than for her colleague. What is the most likely reason?**

- ◯ Salma doesn't speak clearly enough
- ◯ Voices like hers were rare in the recordings it learned from
- ◯ Voice typing never works for anyone
- ◯ Someone set the phone up to be unfair to her

2 **Computers can't be unfair because they're machines.**

◯ True ◯ False

3 **Which of these could cause a data gap?**

- ☐ Testing the tool with many different groups
- ☐ Voice recordings from only a few accents
- ☐ Examples from the people who will use the tool
- ☐ Training photos taken mostly in one country
- ☐ A malaria tool trained mainly on one parasite type

4 **Match each data gap to a fix.**

Voice typing trained on a few accents	• • • •	Test with a balanced set of faces
A malaria tool trained on one parasite	• • • •	Add scams in Kiswahili and other languages
Face tools tested mostly on lighter skin	• • • •	Add recordings of other accents
A scam filter trained only on English	• • • •	Add photos of the other parasite types

5 **Complete the sentence.**

An AI system is only as good as its _____. When some groups are _____ from them, it often makes more _____ for those groups.

Word bank: mistakes · examples · rules · money · missing

Answers · for the teacher

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1 **Voices like hers were rare in the recordings it learned from**

That's a data gap: voices like Salma's were rare in the recordings voice typing learned from, so its best guesses for her are wrong more often.

2 **False**

AI learns from examples that people collect.

3 **Training photos taken mostly in one country; Voice recordings from only a few accents; A malaria tool trained mainly on one parasite type**

Photos taken mostly in one country leave people, places and things from elsewhere rare in the examples; a few accents make the tool err more with other voices; and a malaria tool trained mainly on one parasite type still needs expanding to the others, as its makers at Makerere University say.

4 **Voice typing trained on a few accents — Add recordings of other accents**
A malaria tool trained on one parasite — Add photos of the other parasite types
Face tools tested mostly on lighter skin — Test with a balanced set of faces
A scam filter trained only on English — Add scams in Kiswahili and other languages

Each fix adds or tests the examples that were missing — not just more of the same.

5 **An AI system is only as good as its examples. When some groups are missing from them, it often makes more mistakes for those groups.**

That's the lesson in one line: gaps in the examples often become mistakes for the people left out.

For you to think about

Think of an AI tool your pupils might meet. Who in your class might it work less well for — which languages, accents or faces?